

SWT301 SPRING25 - The Final PE

Online Travel Booking Website

Duration: 80 minutes

Number of Questions: 4

Total Score: 10 points

Instructions:

- Proficiency in at least one programming language is necessary to perform adequately on certain exam questions.
- Students must submit their exam work in a single Word document; no other file formats will be accepted.

Project Context

Liger Travel is developing a new online travel booking website (TravelEase). The system allows users to search for travel destinations, book flights, hotels, and tour packages, and make payments using various methods (credit card, digital wallet, or bank transfer). The system also provides a dashboard for travel agencies to manage bookings, promotions, and customer inquiries.

Project Development Details:

- Project Type: New Software Development
- Development Duration: 12 months
- Development Team: 10 developers
- Testing Team: 5 testers (including 1 test lead)

Constraints:

- The system must handle high traffic, especially during holiday seasons.
- The search and recommendation feature must return accurate and fast results.
- Payment transactions must be secure and error-free.
- The website must be responsive and work across multiple devices.

Question 1 (3 points): Selecting Test Strategy

Based on the project context, choose *one or more* testing strategies from the following:

- Analytical
- Model-based
- Methodical
- Process/Standard-compliant
- Directed (consultative)

- Regression-averse
- Reactive (dynamic)

Answer requirements:

- Identify the most suitable testing strategy for the project.
- Explain why this strategy is appropriate.
- Propose how to implement the strategy in the project.

Question 2 (3 points): Code Review and Bug Identification

Below is a Java method from the Travel Booking website. Identify 6 errors related to coding conventions, coding standards, or coding logic.

```

    public class TravelBooking {
        String bookingId;
        double totalPrice;
        int numberOfTravelers;

        TravelBooking(String id, double price, int travelers) {
            bookingId = id;
            totalPrice = price;
            numberOfTravelers = travelers;
        }

        boolean confirmBooking(String paymentMethod) {
            if (paymentMethod == "CreditCard") {
                return processPayment(totalPrice);
            } else if (paymentMethod.equals("DigitalWallet")) {
                return processPayment(totalPrice * 0.98); // 2% discount
            }
            return false;
        }

        private boolean processPayment(double amount) {
            // Simulate payment processing
            return amount > 0;
        }

        String getBookingDetails() {
            return "Booking ID:" + bookingId + ", Total: " + totalPrice;
        }
    }

```

Answer requirements:

- Identify 6 errors in the above code.
- Explain each error and suggest fixes.

Question 3 (2 points): Component Test Case Design

Below is a method for calculating reward point based on booking amount and customer type.

```
public int calculateRewardPoints(double bookingAmount, String
customerType) {
    if (bookingAmount < 0) {
        return -1;
    } else if (customerType.equals("VIP")) {
        return (int) (bookingAmount * 0.1);
    } else if (customerType.equals("Regular")) {
        return (int) (bookingAmount * 0.05);
    } else {
        return 0;
    }
}
```

Design Unit Test Cases to ensure:

- 100% Statement Coverage
- 100% Branch Coverage
- 100% Equivalence Partitioning Coverage
- 100% Boundary Value Coverage

Answer requirements:

- List all required test cases (including input values and expected output).
- Explain how the test cases ensure full coverage.

Question 4: System Test Case Design (2 points)

Write System Test Cases to test two key business processes in the travel booking system:

1. Booking a Travel Package
2. Searching for Available Hotels

Process 1: Booking a Travel Package

1. User logs into the system.
2. Searches for a travel destination.
3. Selects a travel package (flight + hotel + activities).
4. Reviews package details and price.
5. Enters traveler information.
6. Selects a payment method and completes payment.
7. System confirms the booking and sends a confirmation email.
8. User receives an electronic itinerary.

Process 2: Searching for Available Hotels

1. User visits the website.

2. Enters destination and travel dates.
3. System displays a list of available hotels.
4. User applies filters (e.g., price range, hotel rating, amenities).
5. System updates the search results based on filters.
6. User selects a hotel to view details.
7. User checks availability for selected dates.
8. System displays room availability and booking options.

Answer requirements:

Think about the normal flow, alternative flow, and possible exceptions that may occur in each business process. Write at least 5 test cases for each business process, ensuring you cover:

- **Normal Flow:** The expected process under typical conditions.
- **Alternative Flow:** Variations where users take different actions but still complete the process.
- **Exception Flow:** Error cases where something goes wrong (e.g., invalid input, missing data, system failures).

For each test case, follow the standard format:

1. Test Case ID
2. Test Scenario
3. Preconditions (if applicable)
4. Test Steps
5. Expected Result

Ensure your test cases comprehensively validate the process logic.

Notes: Please feel free to include any assumptions needed for your answers to be clearer and more accurate.